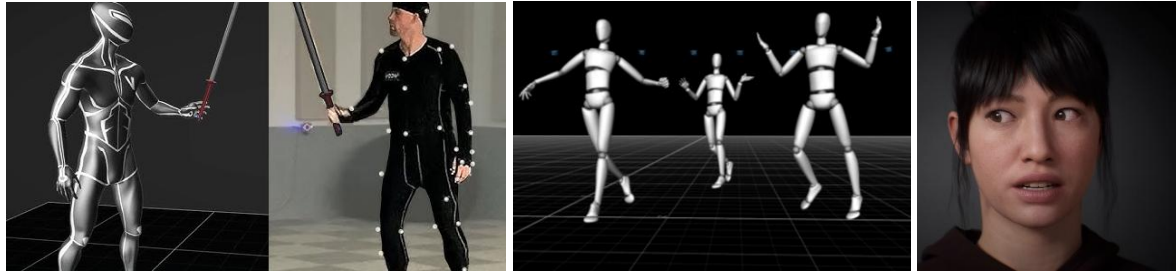


Assignment #04: Motion Editing Demos

Due Date: 17th April 2026

Coursework %: approximately 25%

This assignment aims to showcase your understanding of motion editing, retargeting, and facial animation techniques learned in class. You will work with motion capture data, retarget it to a character, refine the motion using animation tools, and integrate facial animations in Unreal Engine.



Overview

This is an **individual** assignment (no group work).

The objective of this assignment is to demonstrate your understanding of motion capture retargeting, motion editing, and facial animation. You will create three separate demos in Unreal Engine using the provided mocap data: (1) retarget walking animations to multiple MetaHumans with different body types, (2) implement a two-character interaction with user-triggered animation transitions, and (3) showcase facial animation on your own avatar. You should include before-and-after comparisons to highlight the effects of your motion processing and system logic.

You must use Unreal Engine. Assets such as characters and environments can be imported, but the animation should be constructed, and logic should be implemented by you (e.g., you can import a skinned character model, but not one that has full game logic for navigation, you cannot use full environments with props and characters already in them.) Tutorials can be used for guidance but should not be copied directly. Links to any assets used or tutorials followed should be provided in the final report.

Submission Guidelines

1. **Deadline:** Submit your project by 17th April 2026
 - Your submission must include:
 - A **YouTube link** to a 30-60 **second video** showing your demos in action. The video should clearly show the 3 separate demos, demonstrate the features with clear before and after comparisons of motion edits and voice-over or captions.
 - A **PDF report** (example at end) containing:
 - Screenshots of your project, and description of how the retargeting, motion editing and facial animation were applied

- Cite all resources used in your project, including assets, animations, and plugins. Failure to properly cite these sources will result in **0%** for the assignment.
2. **Late Submissions:** Any late submissions will result in **0%** and will be recorded as **absent** for the assignment.
 3. **Original Work:** Ensure that the project is **created by you**. Using someone else's code will be considered cheating.
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Requirements

Your application should be coded in Unreal Engine. Use only **free assets** from the Unreal Engine Marketplace or other free asset libraries (e.g., Metahuman, Mixamo, OpenGameArt). No purchased assets are allowed. You must use **Unreal Engine's animation tools**, such as Sequencer, Character Animation Blueprints, and keyframe animation.

The project should include the following:



1. Demo 1 – Retargeting & Character Diversity (~20%)

- Objective: Showcase how the same motion capture animation can be applied to different MetaHumans.
 - Retarget the provided walking animation to three distinct MetaHumans:
 - Your own avatar (scanned from AR class or create a customized MetaHuman that represents you)
 - A character of a different gender
 - A character with a different body size or age
 - Show all 3 side-by-side in the demo
 - Apply motion edits as needed: foot sliding correction, posture adjustments, warping
 - Higher marks will be awarded for:

- Adapting the motion to suit each character, e.g., adjusting step length, speed, or posture to reflect body size, age, or personality, rather than simply copying the motion
- Critical evaluation in the report
- Document all steps and technical adjustments in your report



2. Demo 2 – Two-Character Interaction (~45%)

- Objective: Build a **plausible interaction** between two MetaHumans controlled by the user.
 - Choose an interaction: rock-paper-scissors, mini fighting game, or dance battle.
 - Implement at least 3 user-triggered animation transitions using a simple UI.
 - Use technical animation tools:
 - Warping for alignment and reach
 - Concatenation of multiple clips
 - Smoothing to remove jitter or abrupt transitions
 - Blend Spaces for smooth motion transitions
 - Keyframe adjustments as needed
 - Two characters must maintain proper alignment during interactions.
 - Higher marks will be awarded for:
 - Smooth, realistic transitions between actions
 - Advanced use of animation tools, e.g., motion matching or dynamic blending
 - Demonstrating thoughtful interaction logic that adapts to user input
 - Document all motion edits, system logic, and workflow in your report.



3. Demo 3 – Facial Animation (~35%)

- Objective: Showcase detailed facial animation on your self-avatar (created with a scan or customized template), using MetaHuman Animator driven by emotional facial video clips.
 - Use MetaHuman Animator to create facial animation from emotional facial video clips downloaded from a high-quality open-source dataset (e.g. [RAVDESS](#))
 - Apply the same clips to a MetaHuman of a different gender or age
 - Create a longer, concatenated sequence combining multiple emotional clips
 - **Higher marks** will be awarded for:
 - Animation quality
 - Comparison of different emotion intensities (e.g. using [MEAD](#) dataset)
 - Sequence & Complexity
 - Critical Evaluation
 - Direct comparison between source video, and both Metahumans
 - Discuss accuracy of emotion transfer and any artifacts
 - Discuss limitations
 - Document settings, workflow, evaluation and logic in your report.

CS7GV5 Report Example

Name:	
Project Title:	
Student ID:	
Youtube link:	

Demo 1: Retargeting & Character Diversity
<i>Screenshot(s) of feature:</i>
<i>Describe your implementation:</i>
<i>Critical evaluation:</i>

Demo 2: Two-character Interaction
<i>Screenshot(s) of feature:</i>
<i>Describe your implementation:</i>
<i>Critical Evaluation:</i>

Demo 3: Facial animation
<i>Screenshot(s) of feature:</i>
<i>Describe your implementation:</i>
<i>Critical Evaluation:</i>

References: Cite all tutorials used, links to all assets used, and Unreal Engine documentation